

# **FAKE AUDIO & SCENE DETECTION USING ACOUSTIC AUDIO & SCENE ANALYSIS**



# OVERVIEW

- Introduction
- Objectives
- DataSets
- Visualization
- Preprocessing
- Model Architecture
- Train & Validation loss over epochs
- Train & Validation Accuracy over epochs

# INTRODUCTION

- **The manipulation of audio recordings “especially in legal or forensic contexts” presents a growing challenge. With the rise of generative models and audio editing tools, distinguishing between real and synthetic audio has become vital. This project investigates deep learning techniques to detect fake or altered audio scenes using state-of-the-art models.**

# OBJECTIVES

- **Detect whether an audio recording is real or fake.**
- **Focus on scene-based manipulation rather than just synthetic speech.**
- **Use pretrained models (AST) to leverage audio spectrogram-based learning.**
- **Evaluate performance across different audio versions and environments.**

# DATASETS

1

## **SceneFake Audio Dataset**

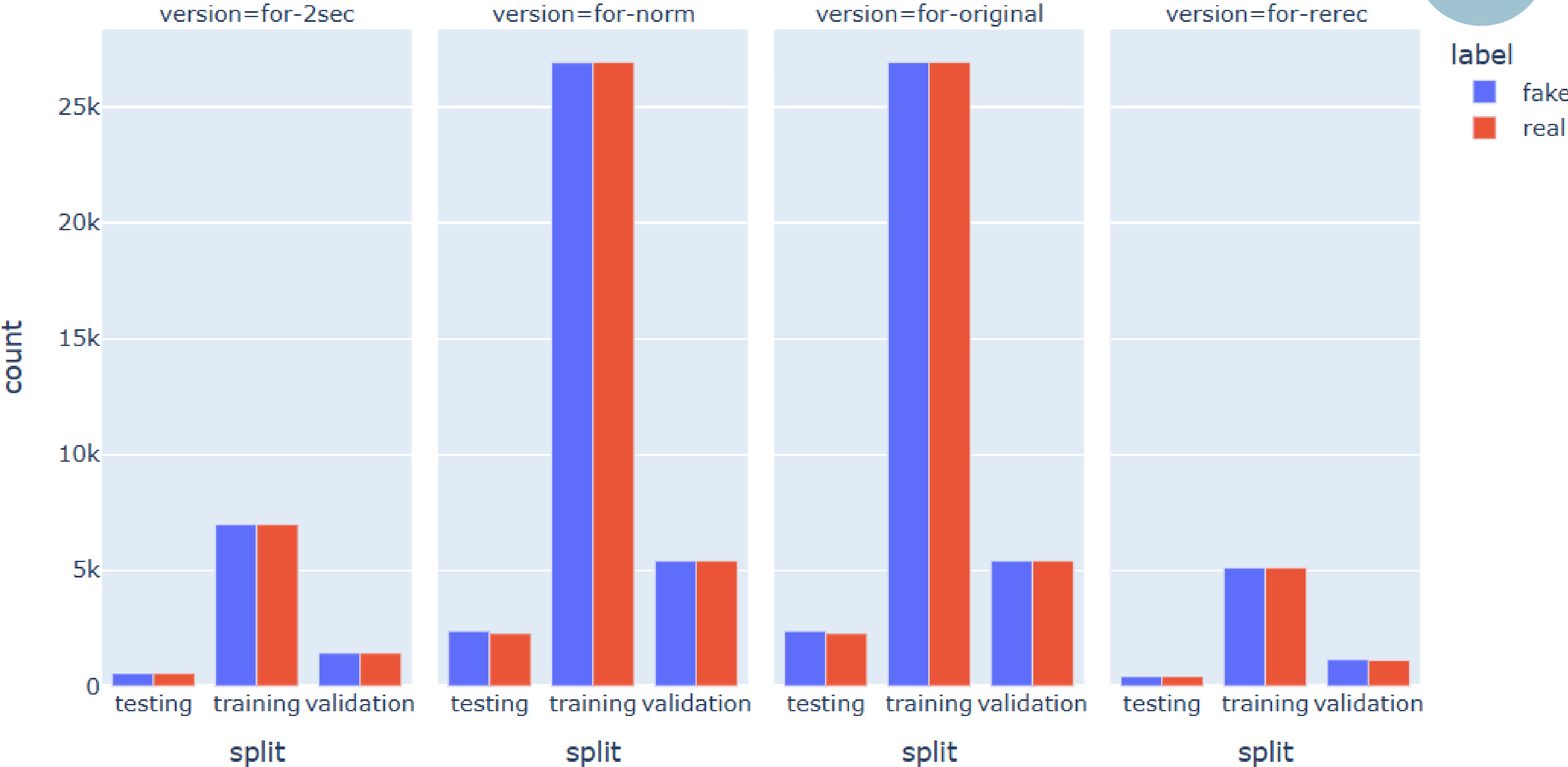
- **~ 60,000 Audio Files (real + fake).**
- **Goal: Detect altered acoustic backgrounds with unchanged speech content.**

2

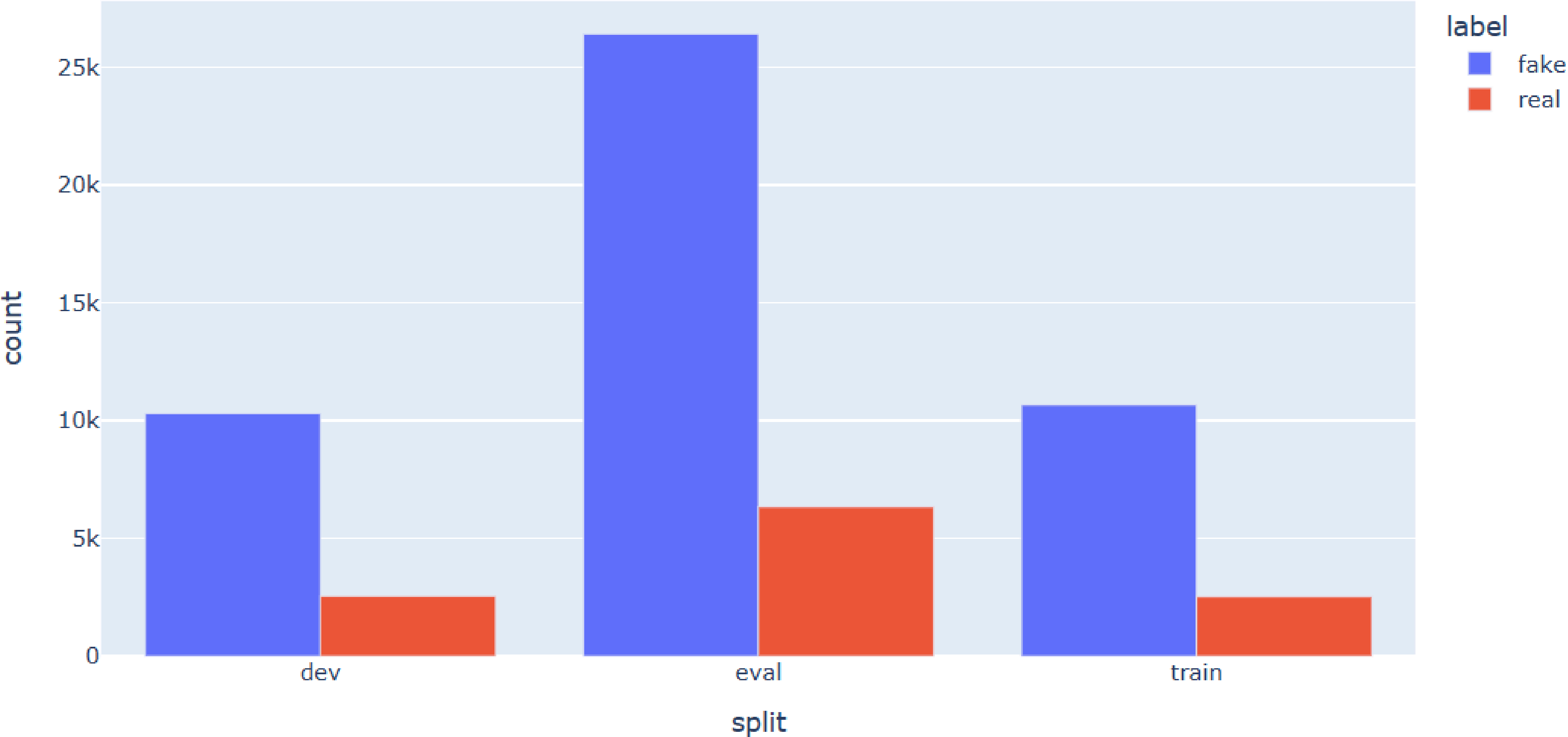
## **Fake-or-Real Audio Dataset**

- **~170,000 Audio files (real + fake).**
- **Goal: To differentiate between human voice and generated voice by Ai tool.**

# Number of Real and Fake Audios per Split and Version

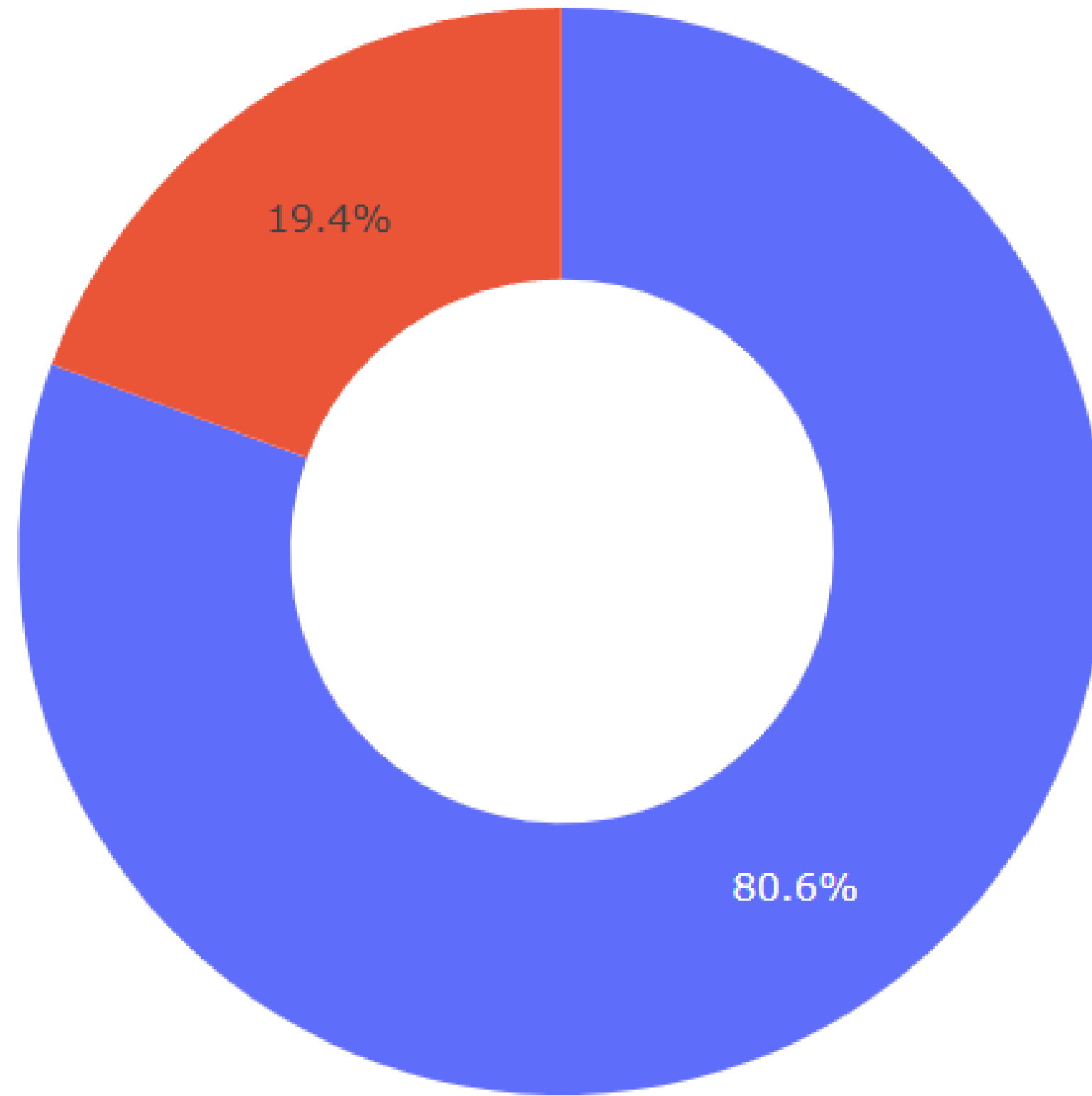


Number of Real and Fake Audios per Split in SceneFake Dataset



## Label Distribution in SceneFake Dataset

8



fake  
real

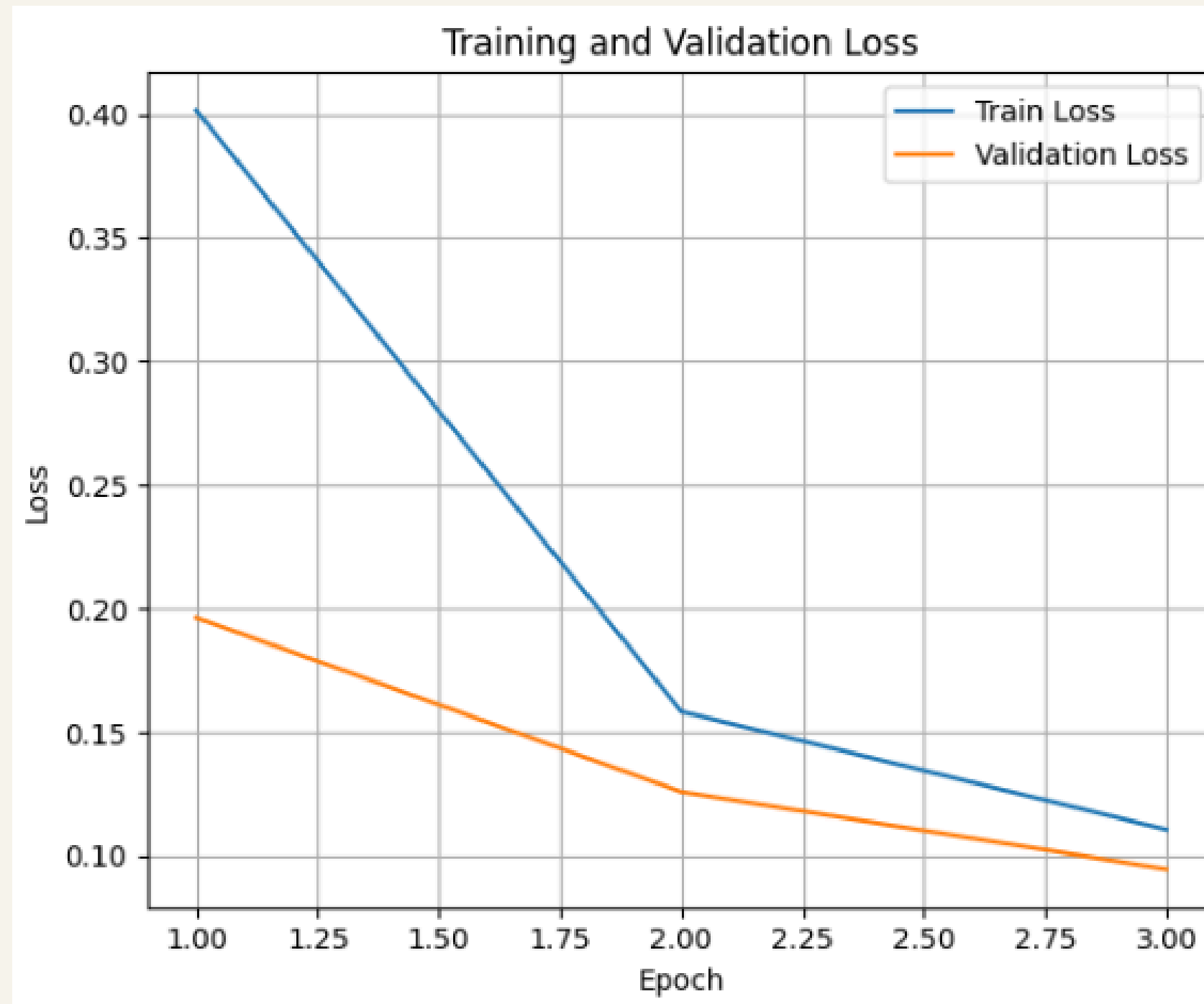
# PREPROCESSING

- Merge two datasets.
- Ensure to Convert all audio files from mp3 to wav.
- Ensure all audio files are valid (Not empty / damage)
- All audio files trim or padded to 5 sec.
- Converted audio to spectrograms (AST Feature Extractor).
- Used Label Encoding for categorical labels.
- Balanced datasets and cleaned metadata .
- Split (train/val/test) and label ( scene real/ scene fake/ voice real / voice fake ).

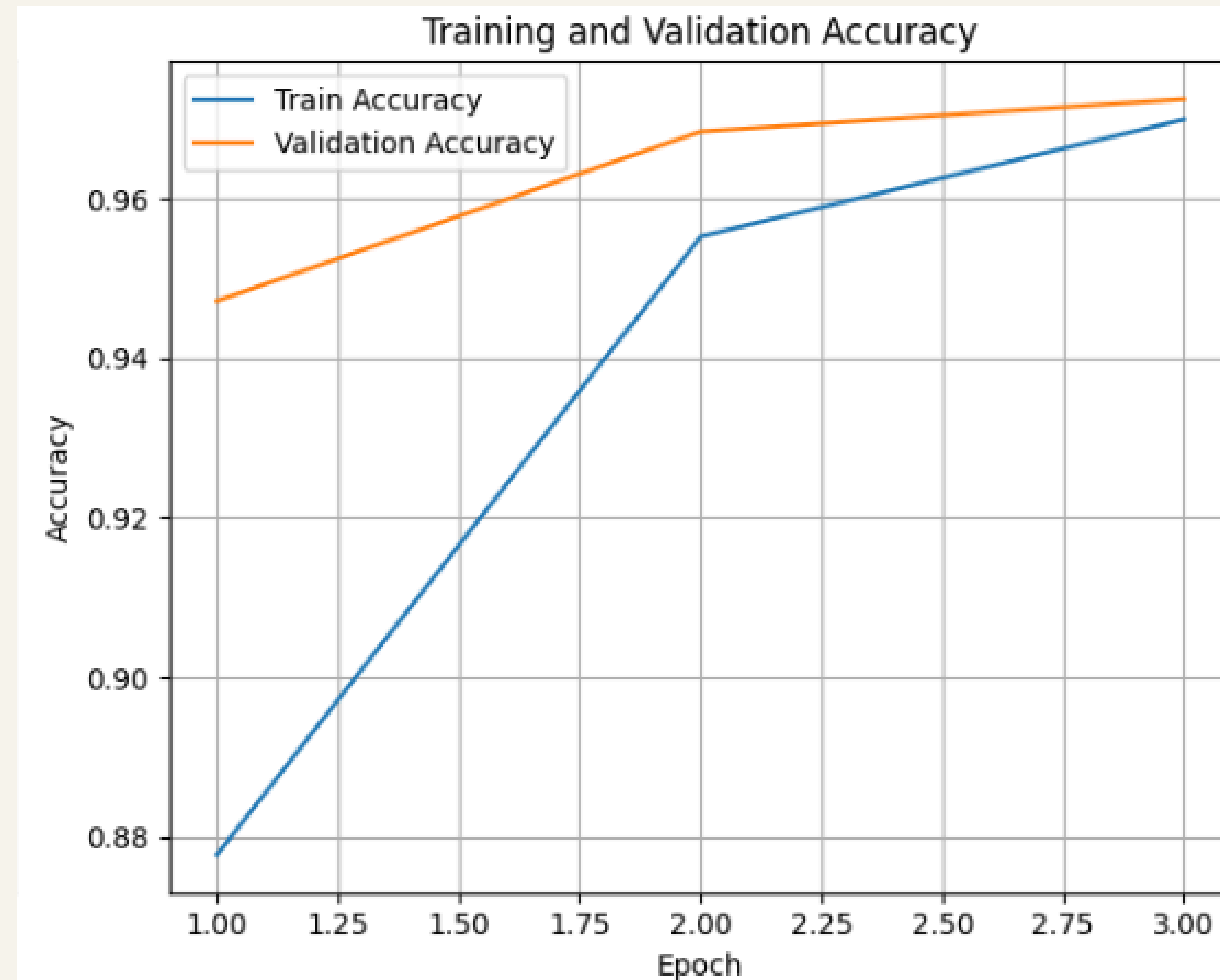
# MODEL ARCHITECTURE

- Utilized AST (Audio Spectrogram Transformer) from Hugging Face Transformers.
  - Loaded with AST FeatureExtractor and AST For AudioClassification.
  - Configured to fine-tune on multi class classification .
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# TRAIN & VALIDATION LOSS OVER EPOCHS



# TRAIN & VALIDATION ACCURACY OVER EPOCHS





**THANK YOU**